Introduction to Machine Learning

CSDSS105: Machine Learning

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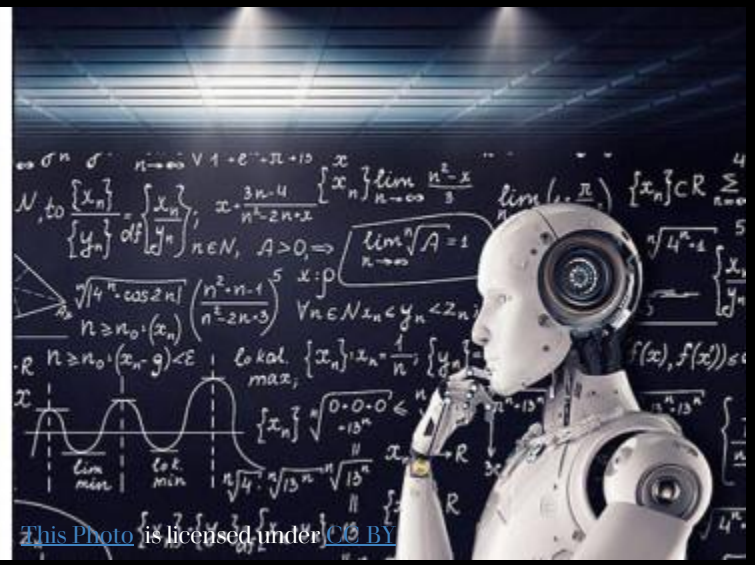

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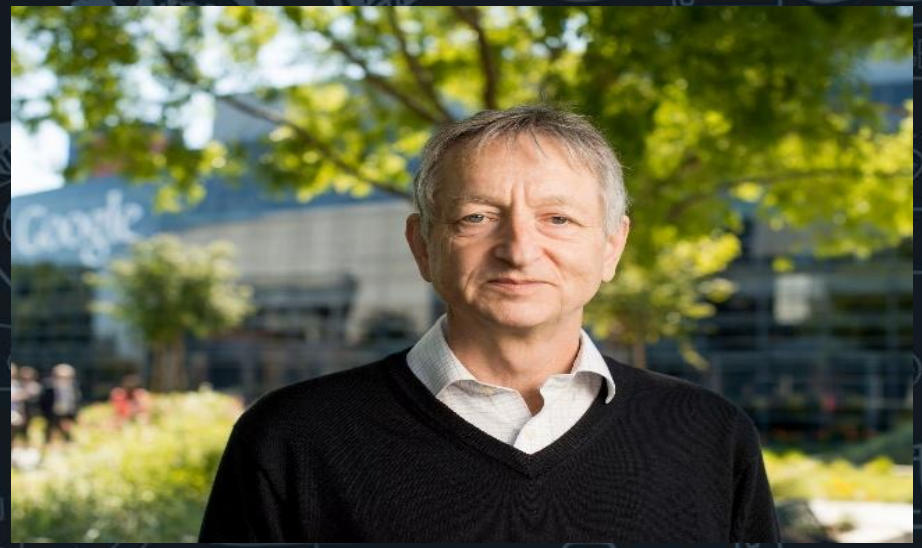
Machine Learning Overview

The godfather of AI, Geoffrey Hinton, in 2024 shared a Nobel Prize in Physics for his contribution to ML

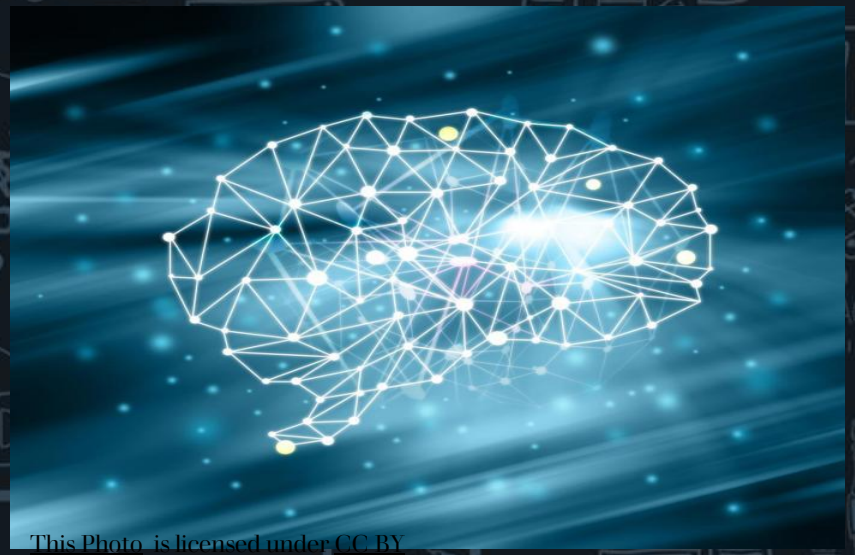
He has published a paper showing how to train a deep neural network capable of recognizing handwritten digits with state-of-the-art precision (>98%). They branded this technique “deep learning”.



Deep learning is a simplified model of our cerebral cortex, composed of a stack of layers of artificial neurons.

Training a deep neural network was widely considered nearly impossible in the late 1990s without the help of tremendous computing power and great amounts of data

A decade later, machine learning had conquered the industry, and today it is at the heart of much of the magic in high-tech products, ranking your web search results, powering your smartphone’s speech recognition, recommending videos, and perhaps even driving your car.



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Objectives and Approach

I assume that you know close to nothing about machine learning. The goal is to give you the concepts, tools, and intuition you need to implement programs capable of learning from data.

The course is designed:

- Learn most of the Mathematical foundations of Machine Learning (ML)
- No heavy proofs, focus on understanding and implementation
- Gradual transition from Math → ML concepts → applications
- Learn many ML techniques

In this course we are going to use production-ready Python frameworks:

➢ Scikit-Learn: it implements many machine learning algorithms efficiently

Prerequisites

- Some programming experience. At least, you are familiar with Python's main scientific libraries, in particular, NumPy, Pandas, and Matplotlib.
- Linear Algebra will help you understand how machine learning algorithms work (not just how to use them).
- Some concepts from Multivariate calculus, calculus, probability, and statistics

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What is Machine Learning?

Machine learning is the science (and art) of programming computers so they can learn from data.

A computer program is said to learn from experience E with respect to some task T and some performance measure P, if its performance on T, as measured by P, improves with experience E [Tom Mitchell, 1997]

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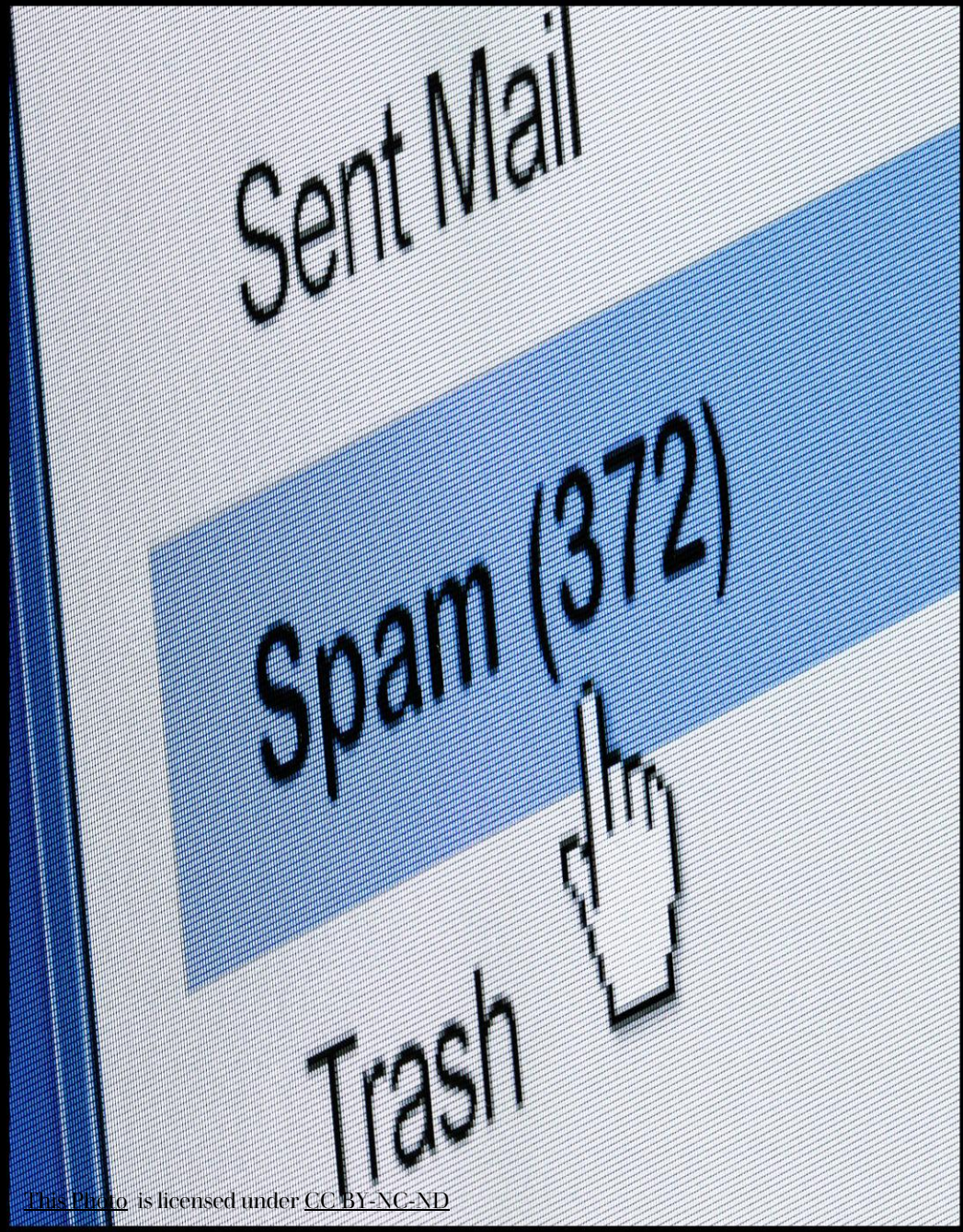
Example of Machine Learning Program:

Your spam filter is a machine learning program that, given examples of spam emails (flagged by users) and examples of regular emails (nonspam, also called “ham”), can learn to flag spam.

- ❑ The examples that the system uses to learn are called the **training set.**
- ❑ Each training example is called a **training instance** (or **sample**).
- ❑ The part of a machine learning system that learns and makes predictions is called **a model.**

To reflect this example to the definition by Mitchell,

The task *T* is to flag spam for new emails, the experience *E* is the training data, and the performance measure *P* needs to be defined; for example, you can use the ratio of correctly classified emails. This particular performance measure is called **accuracy**, and it is often used in classification tasks.



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Why Use Machine Learning?

To understand why us ML, let us consider how you would write a spam filter using traditional programming, see figure 1-1

🔍 **Identify spam characteristics**
Look for common keywords like “4U,” “credit card,” “free,” and “amazing” in subject lines, as well as patterns in sender names and email content.

🧠 **Develop detection algorithms**
Create algorithms that recognize each pattern you've identified.

🚩 **Flag suspicious emails**
Mark emails as spam when multiple patterns are detected.

🔁 **Test and refine**
Continuously test the program and adjust the detection logic until performance is satisfactory.

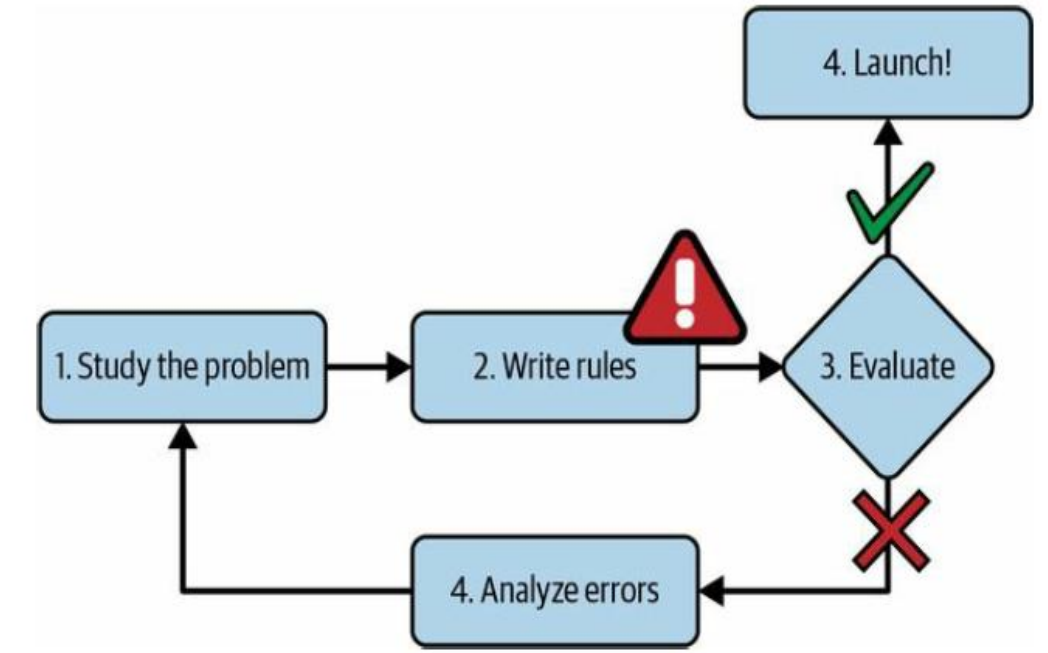


Figure 1-1. The traditional approach

Bad side

The program will likely become a long list of complex rules, pretty hard to maintain.

In contrast, a spam filter based on machine learning techniques automatically learns which words and phrases are good predictors of spam by detecting unusually frequent patterns of words in the spam examples compared to the ham examples (Figure 1-2). The program is much shorter, easier to maintain, and most likely more accurate.

What if the spammer understands and, because of that, he changes the keywords?

Example: spammer notices that all their emails containing "4U" are blocked

A spam filter using traditional programming techniques would need to be updated to flag "For U" emails. If spammers keep working around your spam filter, you will need to keep writing new rules forever.

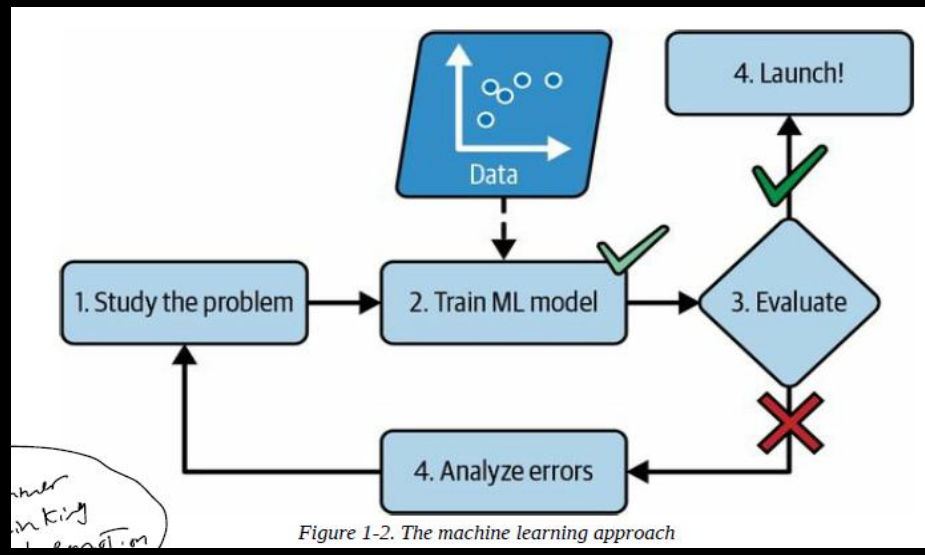
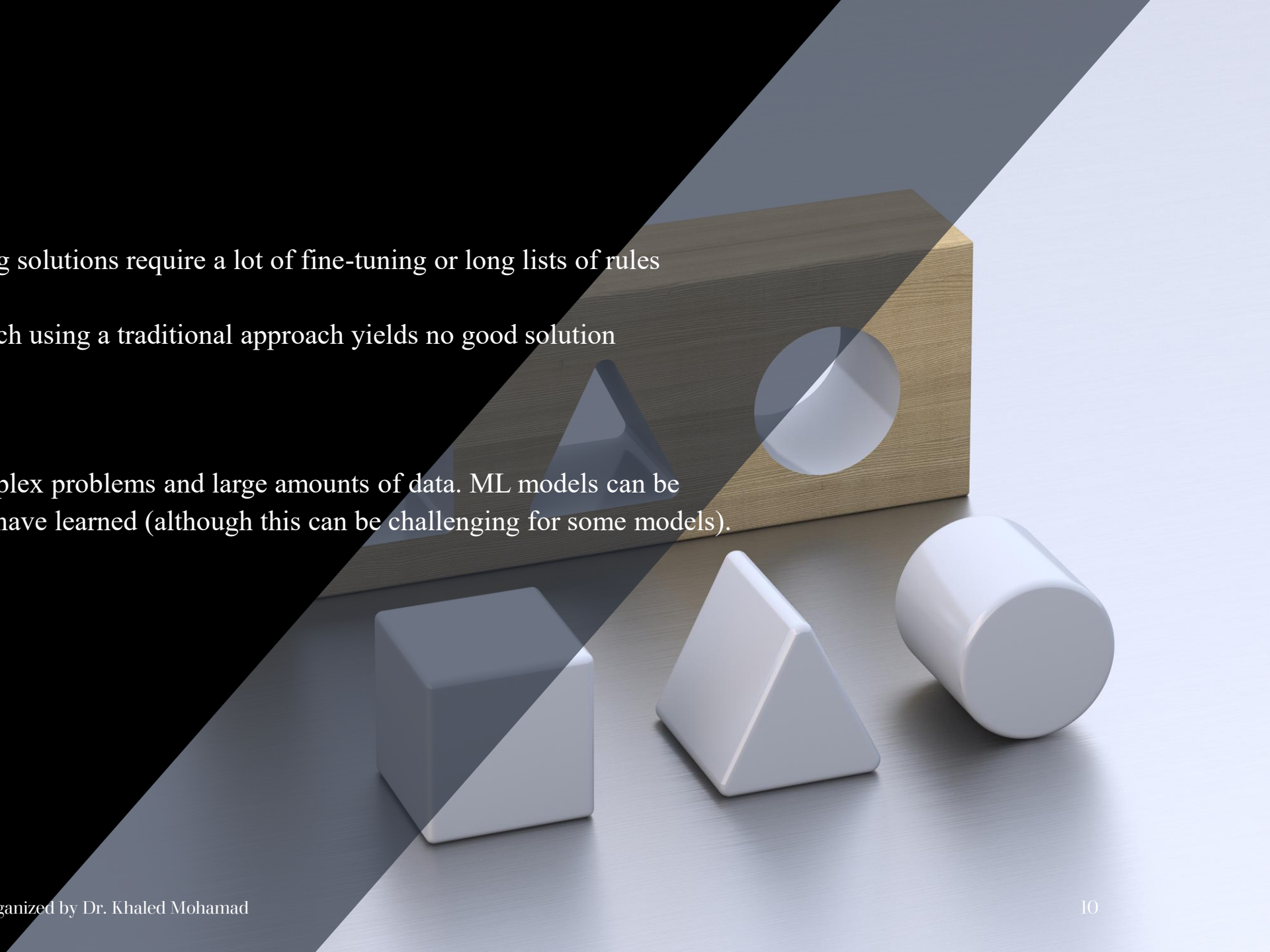


Figure 1-2. The machine learning approach

In contrast, a spam filter based on machine learning techniques automatically notices that "For U" has become unusually frequent in spam flagged by users, and it starts flagging them without your intervention.

ML is great for

- Problems for which existing solutions require a lot of fine-tuning or long lists of rules
- Complex problems for which using a traditional approach yields no good solution
- Fluctuating environments.
- Getting insights about complex problems and large amounts of data. ML models can be inspected to see what they have learned (although this can be challenging for some models).



Examples of Applications

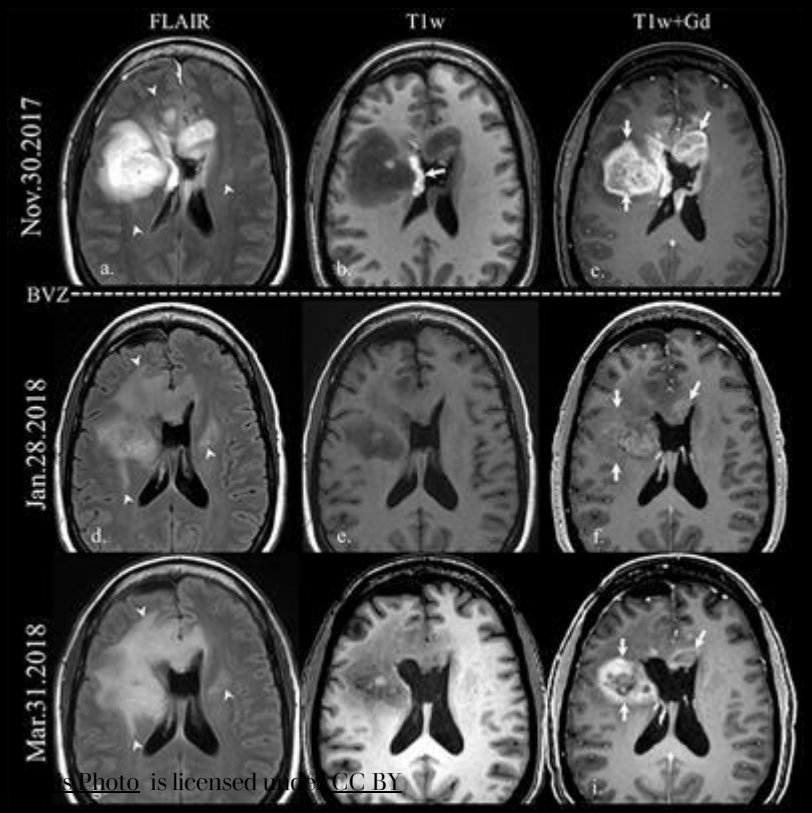
To give you a sense of the incredible breadth and complexity of the tasks that machine learning can tackle, and the types of techniques that you would use for each task., let's look at some concrete examples of machine learning tasks:

➢ Analyzing images of products on a production line to automatically classify them

This is image classification, typically performed using convolutional neural networks (CNNs) or sometimes transformers

➢ Detecting tumors in brain scans

This is semantic image segmentation, where each pixel in the image is classified (as we want to determine the exact location and shape of tumors), typically using CNNs or transformers.



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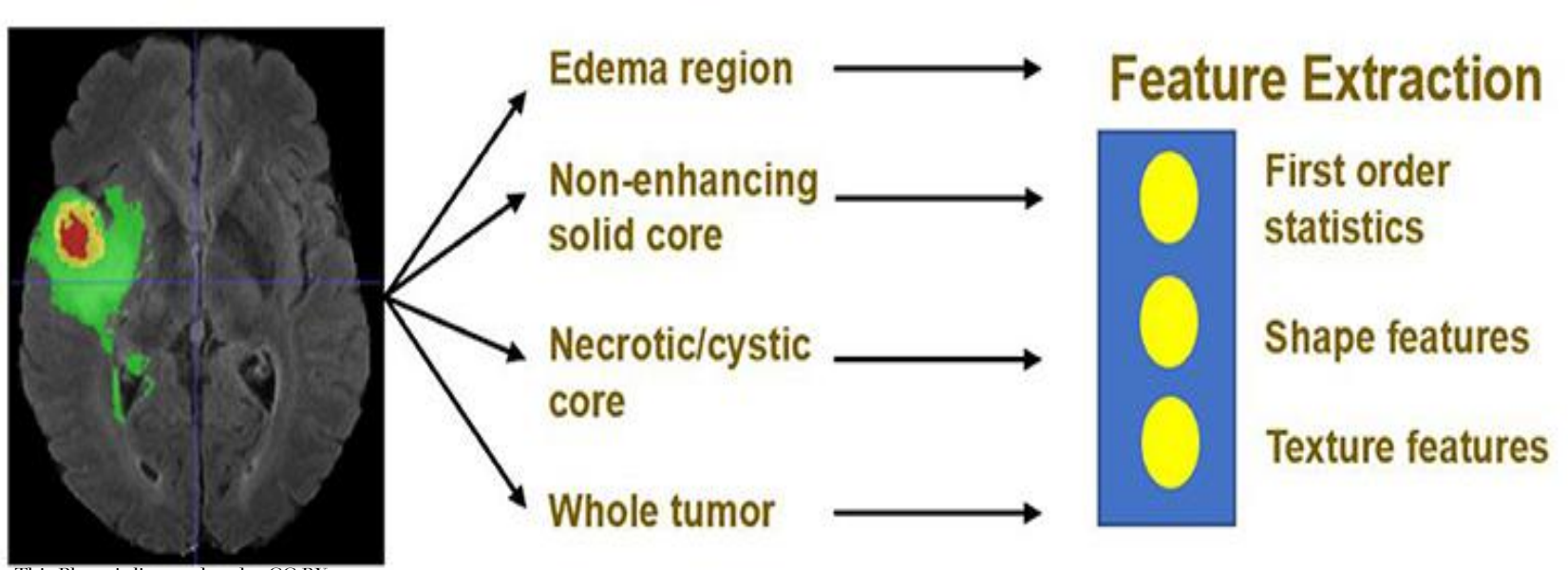

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➢ Automatically classifying news articles OR automatically flagging offensive comments on discussion forums

This is natural language processing (NLP), and more specifically text classification, which can be tackled using recurrent neural networks (RNNs) and CNNs, but transformers work even better

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➢ Making your app react to voice commands

This is speech recognition, which requires processing audio samples. We can use RNNs, CNNs, or transformers.

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- **Segmenting clients based on their purchases so that you can design a different marketing strategy for each segment**

This is clustering, which can be achieved using *k*-means, DBSCAN, and more

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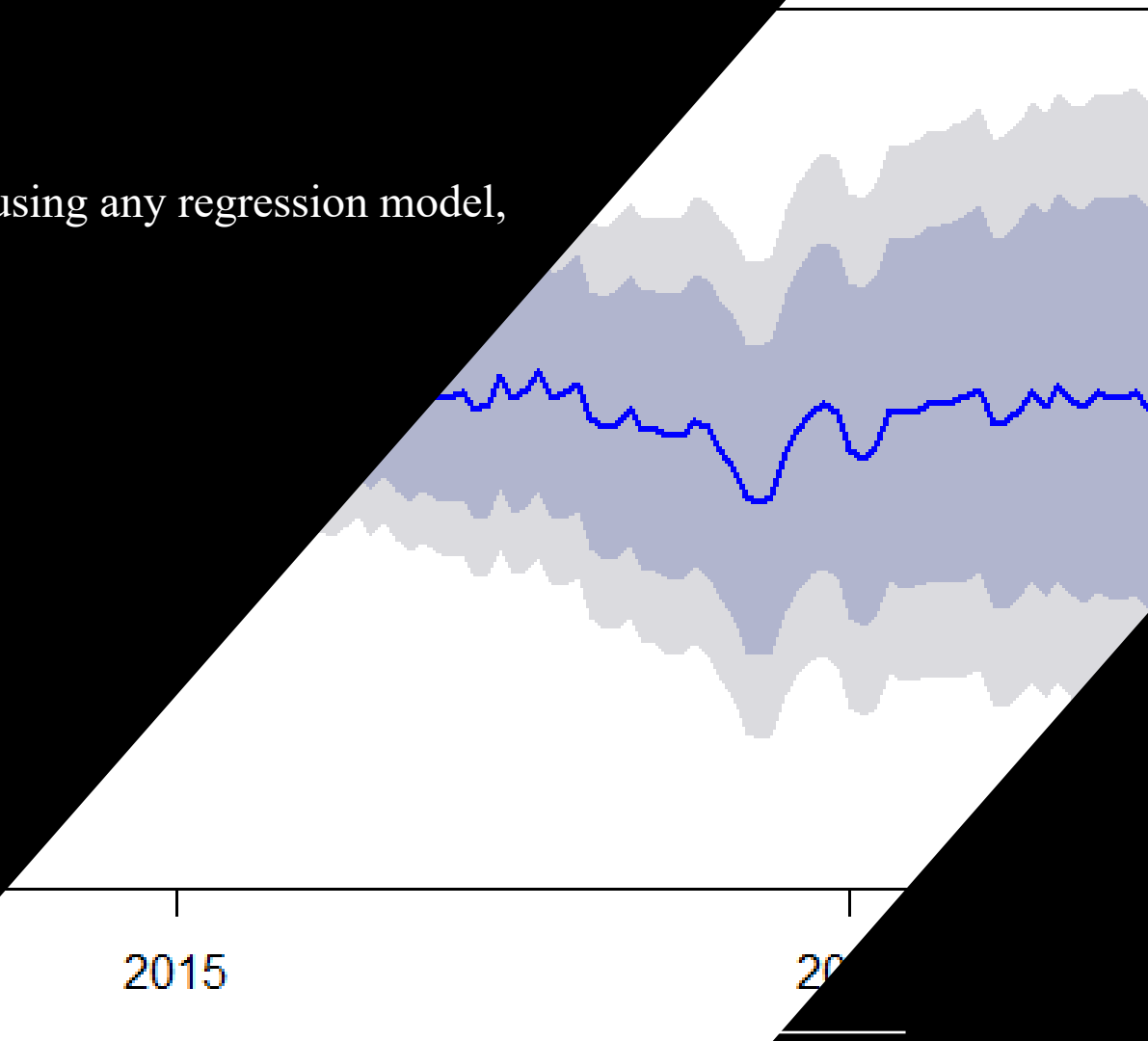
➢ Forecasting your company's revenue next year, based on many performance metrics.

This is a regression task (i.e., predicting values) that may be tackled using any regression model, such as a linear regression or polynomial regression mode



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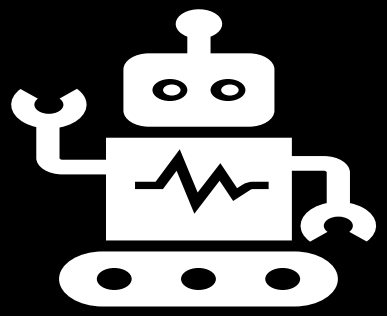
➢ Recommending a product that a client may be interested in, based on past purchases

This is a recommended system. One approach is to feed past purchases (and other information about the client) to an artificial neural network and get it to output the most likely next purchase. This neural net would typically be trained in past sequences of purchases across all clients.

Types of Machine Learning Systems



There are so many different types of machine learning systems that it is useful to classify them in broad categories, based on the following criteria:

- **Training Supervision**: How they are supervised during training (supervised, unsupervised, semi-supervised, self-supervised, and others)

  Supervision in ML refers to the type of information (label or guidance) provided to the algorithm during training. It describes how much the algorithm knows about the correct answers while learning from data

- **Online versus batch learning**: Whether or not they can learn incrementally on the fly

- **Instance-based versus model-based learning**: Whether they work by simply comparing new data points to known data points, or instead by detecting patterns in the training data and building a predictive model, much like scientists do.

An example of describing an ML system, an online, model-based, supervised learning system.

Spam filter may learn on the fly using a deep neural network model trained using human-provided examples of spam and ham.

Training Supervision

ML systems can be classified according to the **degree and form of guidance (labels, signals, or feedback)** the model receives during training.

- To simplify the types based on supervision, think about supervision as a teacher guiding students. Sometimes the teacher gives you all the answers, sometimes only a few, and sometimes you figure things out by yourself.

Here are the main types:

1- Supervised learning (Learning with a teacher)

The model is trained with both questions (inputs) and answers (labels). The goal is to learn the relationship between inputs and outputs, so you learn the rules that connect them. Therefore, it can predict answers for new questions.

Examples:

- Predicting house prices: input /features are size, location, and number of rooms. Output / target is price.
- Classify whether an email is spam (Email filtering): : input /features is email text. Output / target is “spam “ or “not spam”
- Recognize an image (Image recognition): Input is picture. Output = “cat” or “dog”

2- Unsupervised Learning (Learning without a teacher)

The model only sees questions (inputs) but no answers (labels). It must find patterns or groups on its own. The goal is to discover hidden structures in data.

Examples:

- Customer segmentation: Grouping shoppers into “bargain hunters,” “loyal customers,” etc., without knowing labels beforehand.
- Document clustering: Grouping news articles into politics, sports, or health without predefined categories.
- Market basket analysis: Discovering that people who buy bread often buy butter too.